

Team Name: Green Spark

Project Name: PlantAid

Maker Fair

SDG 2 – Zero Hunger

Team Members



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Acknowledgement Remarks (If Applicable)

Problem Statement:

- **Agricultural crops are highly vulnerable to diseases that often go undetected at early stages, leading to significant yield loss, food insecurity, and economic hardship for farmers. Traditional disease diagnosis is manual, time-consuming, and often requires expert knowledge, which is not accessible to all farmers, especially in rural areas. There is a critical need for an affordable, accessible, and accurate solution to help farmers identify plant diseases early and take timely action to protect their crops.**

Introduction:

- **PlantAid is an AI-powered solution designed to detect plant diseases using just an image of a leaf. It utilizes a Convolutional Neural Network (CNN) built with TensorFlow to classify diseases with high accuracy. The platform is supported by a user-friendly web interface, enabling farmers and agricultural workers to upload leaf images and receive instant diagnoses—no expert knowledge required. By combining machine learning with accessible technology, PlantAid aims to support sustainable farming, reduce crop loss, and empower farmers with actionable insights.**

Research Survey:

- To ensure PlantAid addresses real agricultural needs, a focused research survey was conducted through the following approaches:
- **Data Collection:** Publicly available plant disease image datasets were collected from Kaggle to train the model. Supplementary leaf images were gathered from plants across our university campus to increase dataset diversity.
- **Field Insights:** Conversations were held with local farmers to understand their current challenges in identifying and managing plant diseases. Most farmers lack access to expert diagnosis and depend on experience or trial-and-error, often resulting in delayed treatment and crop damage.
- **Conclusion:** The findings confirmed a strong demand for an automated, easy-to-use, and accessible plant disease detection system, especially in rural and resource-limited settings.

Proposed Solution:

- **"PlantAid" uses a Convolutional Neural Network (CNN) model to detect plant diseases by analyzing images of plant leaves. Built with TensorFlow, the system classifies leaf images into healthy or diseased categories, enabling early disease detection. This provides farmers with a fast, accessible tool to prevent crop loss and take timely action. The solution is easy to use, requires no technical expertise, and leverages AI to improve agricultural productivity.**

Prototype:

- The "PlantAid" prototype is accessible via the following URL: <https://plant-aid-client.vercel.app> . It allows users to upload images of plant leaves for disease detection. The system uses a TensorFlow-based Convolutional Neural Network (CNN) to analyze the images and return accurate results on whether the plant is healthy or diseased.
- Key features of the prototype:
 - Image Upload: Users can upload leaf images through a simple interface.
 - Disease Detection: The CNN model processes the image and classifies the plant's condition.
 - Diagnosis Results: The system displays the detected disease and offers treatment recommendations.
 - User-Friendly Interface: Designed for easy navigation without any technical expertise required.

Advantages & Disadvantages:

- **Advantages:**

1. **Early Disease Detection:** Helps farmers prevent crop loss by identifying diseases early.
2. **Efficient:** Saves time and resources compared to manual inspections.
3. **User-Friendly:** Easy to use, no technical skills required.
4. **Scalable:** Can recognize more diseases and add features.
5. **Accessible:** Can be used on mobile/web, reaching a wide audience.

- **Disadvantages:**

1. **Image Quality Dependent:** Accuracy relies on the quality of the uploaded images.
2. **Limited Dataset:** Can only detect diseases in the trained dataset.
3. **Internet Access Needed:** Requires a stable internet connection for image upload and processing.

Conclusion:

- **"PlantAid" offers an innovative solution to help farmers detect plant diseases early, ensuring better crop management and reducing losses. By leveraging AI and machine learning, the system provides a fast, efficient, and user-friendly way to diagnose plant diseases through image recognition. While the prototype shows great potential, future enhancements, including expanding the disease dataset and improving image quality handling, will further strengthen its effectiveness. "PlantAid" is a step toward smarter, more sustainable agriculture.**

References:

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Thank You

